Polynomial Functions

Rail Engineering Mathematics

Sheffield Hallam University

Lecture 1

Learning Outcomes

- Recognise polynomials.
- Recognise typical shapes of polynomial graphs (constant, linear and quadratic).
- Interpret linear equations.
- Solve quadratic equations.
- Plot polynomial graphs using MATLAB.

Function notation

A **function** tells us how one variable depends on (possibly multiple) others. For example, $y = 3x^2 + 8x - 7$. Here the **independent variable** is x and the **dependent variable** is y; we say that y is dependent upon x. That is, the value of y depends on the value of x that we put in.

We could also express the function as:

$y(x) = 3x^2 + 8x - 7$,

$f(x) = 3x^2 + 8x - 7$,

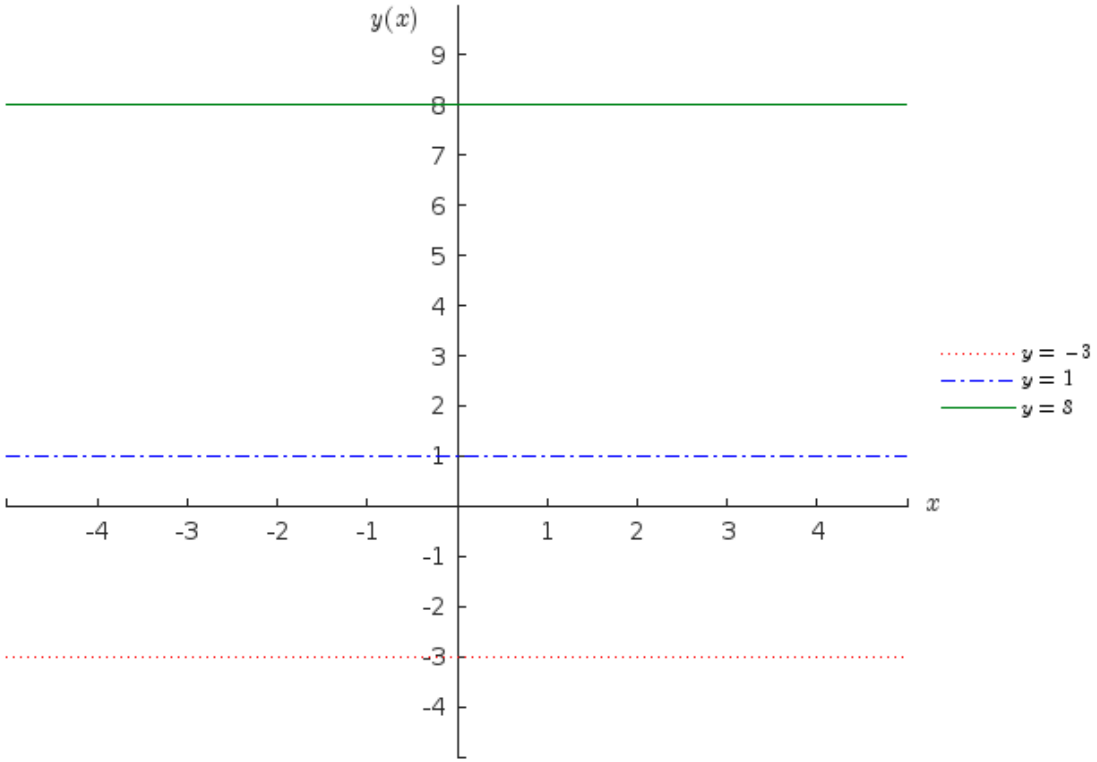
$g(x) = 3x^2 + 8x - 7$, etc

Here the independent variable is explicitly x, and the function is named y, f, or g.

Polynomials

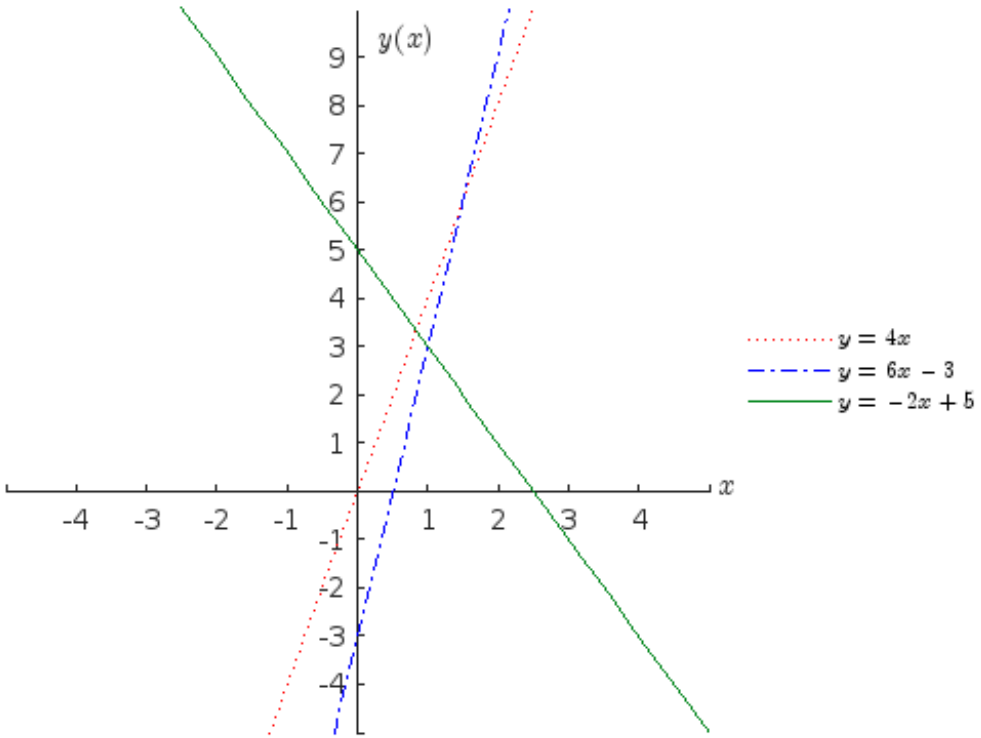
- A **polynomial function** is one that *only* involves non-negative integer powers of x, for example:
  - $y = 7x + 4$ (polynomial of order/degree 1, linear)
  - $y = 3x^2 - 5x - 1$ (polynomial of order/degree 2, quadratic)
  - $y = -x^3 + 5x^2 - 7x + 12.01$ (poly. of order/degree 3, cubic)
- Functions containing negative or non-integer powers, or other functions, (such as trigonometric) are *not* polynomials, e.g.
  - $y = x^2 + 4\sqrt{x} - 5$
  - $y = \dfrac{5}{x^2} - 7x^3 + 6x - 4$
  - $y = x^2 + \sin(x)$

Graphs of Polynomials: Constant functions



Graphs of **constant functions** (no dependency on x) are always straight, horizontal lines.

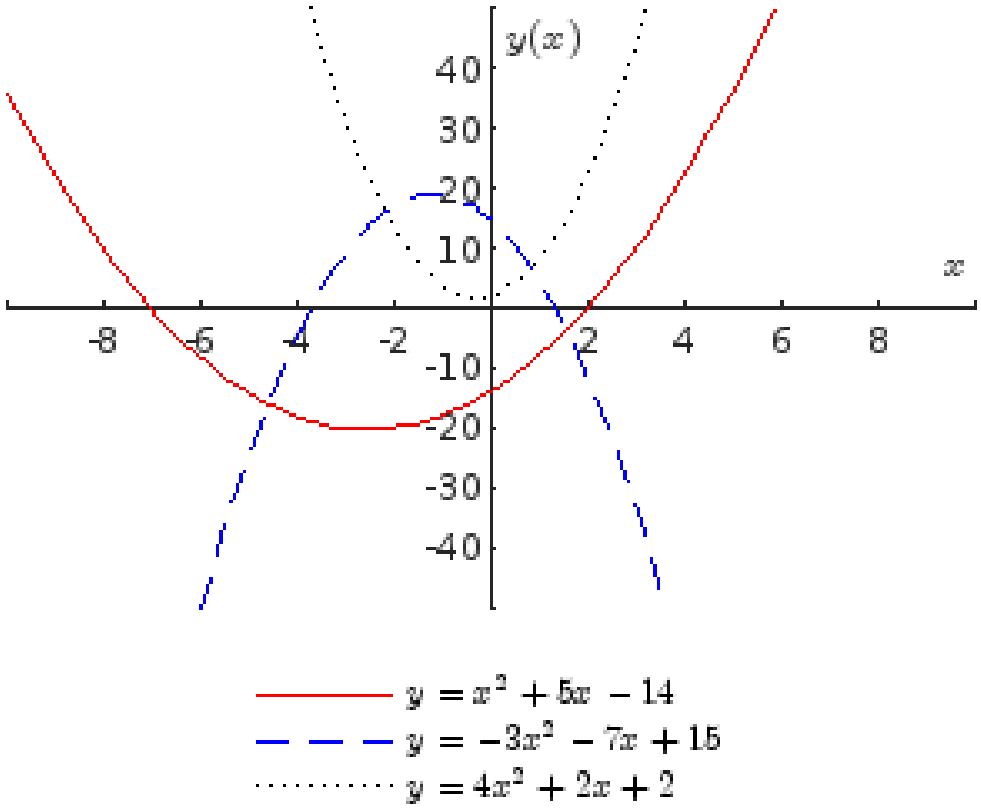
Graphs of Polynomials: Linear functions



Graphs of **linear functions** are always straight lines.

Graphs of Polynomials: Quadratic functions

Quadratic curves are either $\cup$ or $\cap$-shaped depending on the values of a.



Graphs of Polynomials: Quadratic functions

When trying to visualise a quadratic function, consider:

- What is the orientation?
  - $a > 0$: upturned.
  - $a < 0$: downturned.
- Is it broad or narrow compared to $y = x^2$?
  - $|a| > 1$: narrower.
  - $|a| < 1$: broader.
- Where is the turning point?
  - Positive c will push it up.
  - Negative c will push it down.
  - Positive or negative b will push it down if $a > 0$ (up if $a < 0$).
  - $b = 0$: on the y-axis.
  - $b > 0$: left of the y-axis if $a > 0$ (right if $a < 0$).
  - $b < 0$: right of the y-axis if $a > 0$ (left if $a < 0$).

Determining the Linear Equation

Equation of a straight line

$$y = mx + c$$

represents a straight line, where m and c are constants.

m is the **gradient** (slope) of the line and can be calculated as

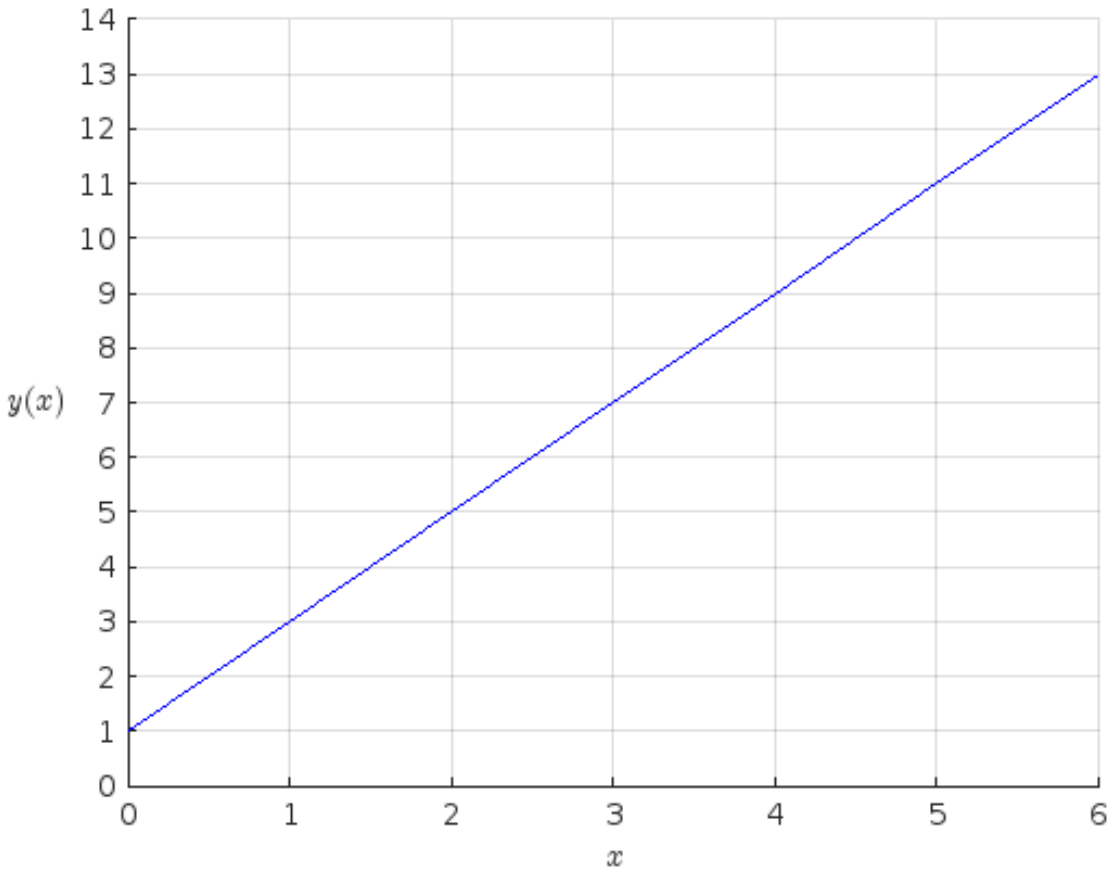
$$m = \frac{\text{vertical change}}{\text{horizontal change}} = \frac{\Delta y}{\Delta x}$$

c is the value of y when the line crosses the y-axis (at $x = 0$), known as the y-**intercept**.

Note: to find where the line crosses the x-axis, simply let $y = 0$.

Determining the Linear Equation

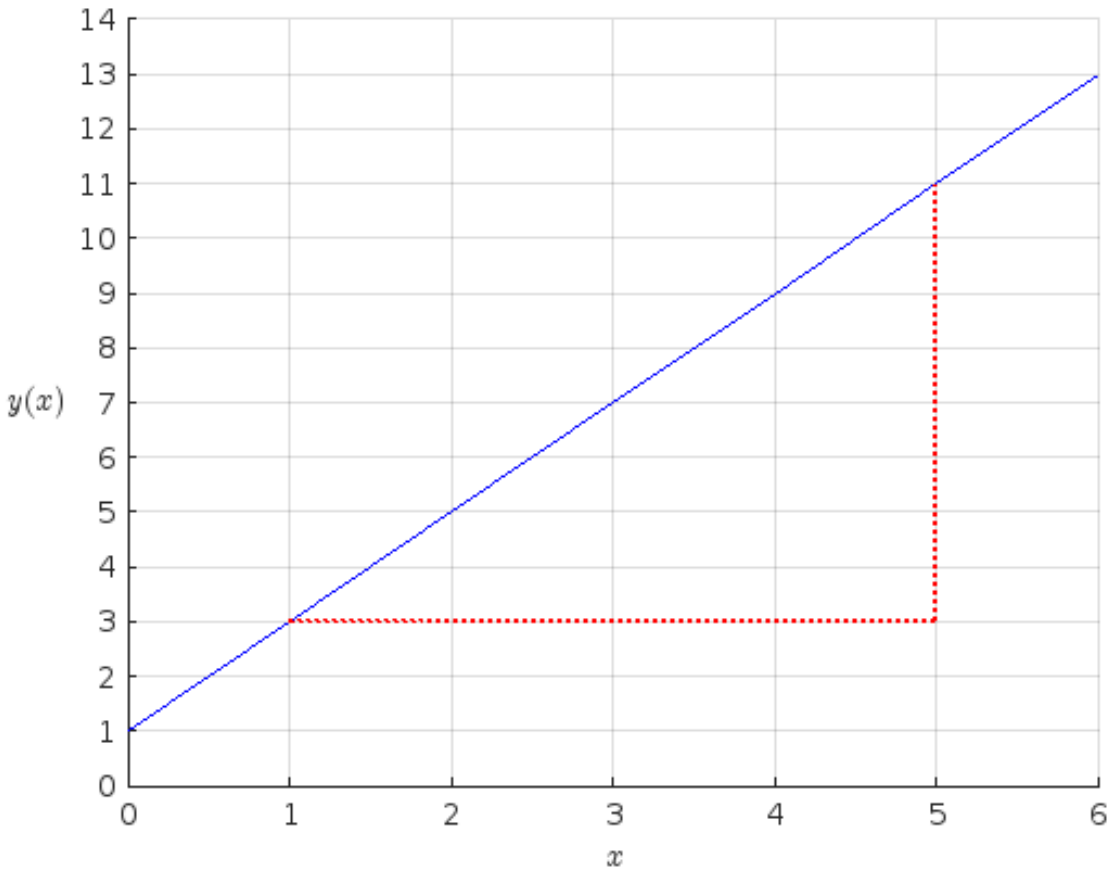
Example: Find the equation of this line:



Firstly, we can see that $c = 1$, as this is the height where the y-axis is crossed.

Determining the Linear Equation

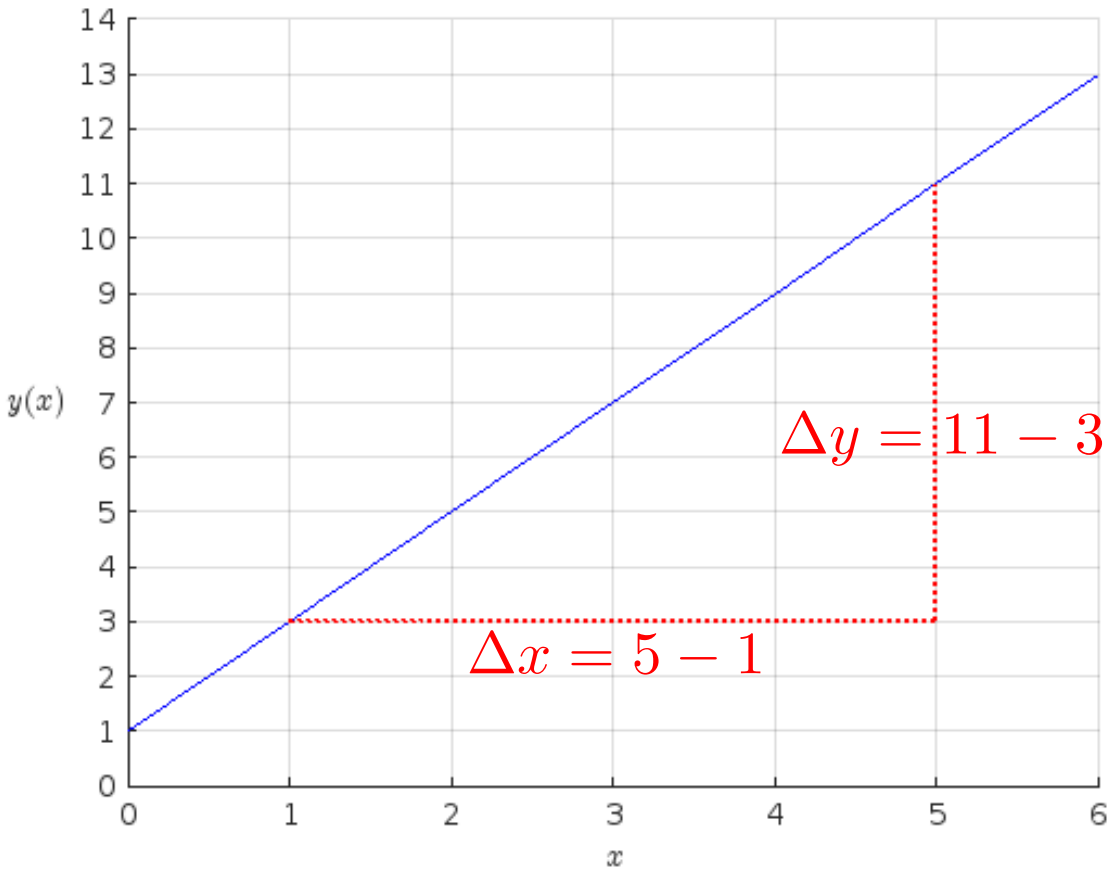
Example: Find the equation of this line:



Secondly, we can determine the gradient m by constructing a triangle on the graph.

Determining the Linear Equation

Example: Find the equation of this line:



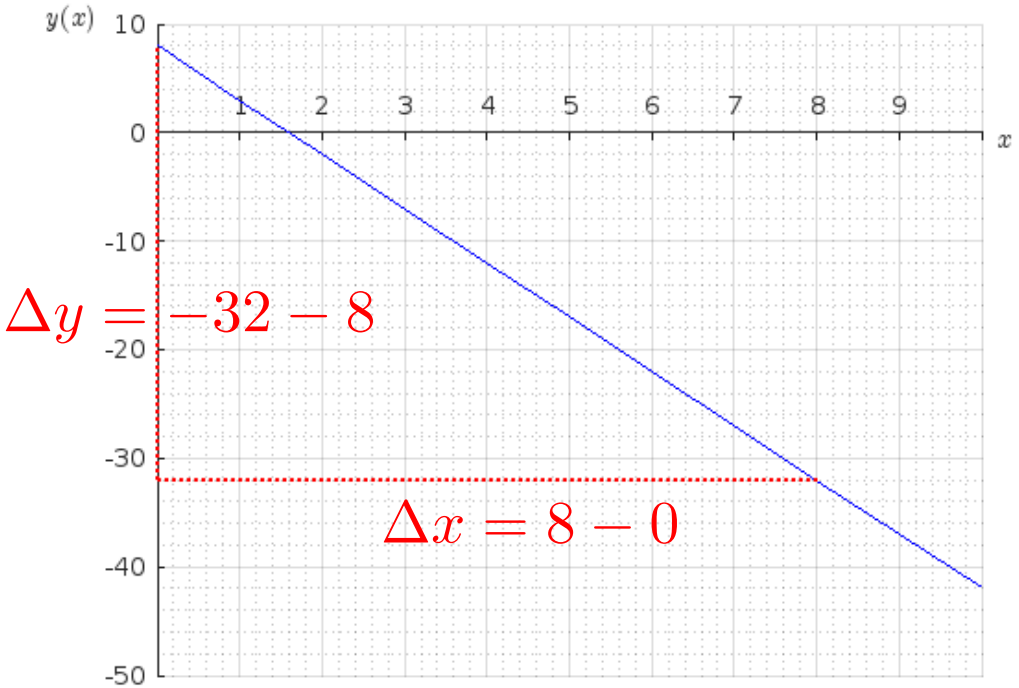
$$m = \frac{\Delta y}{\Delta x} = \frac{8}{4} = 2$$

$$\therefore \quad y = 2x + 1$$

Determining the Linear Equation

Note that if the straight line graph is *decreasing* then we expect a **negative gradient**.

This is because there is a decrease in the value of y as x increases.



This time the gradient is negative as there is a decrease from left to right:

$$m = \frac{\Delta y}{\Delta x} = \frac{-40}{8} = -5$$

$$\therefore \quad y = -5x + 8$$

Quadratic Functions

All quadratics can be represented in the general form:

The quadratic equation

$$y = ax^2 + bx + c,$$

where a, b and c are constants and $a \neq 0$.

Quadratic Functions

The constant c is the y-intercept, as in the linear case (if $x = 0$ then the equation becomes $y = a \times 0^2 + b \times 0 + c = c$).

The curve can cross the x-axis (at $y = 0$) twice, once (just touching it) or never. If it does cross the x-axis, we can calculate the values of x where this occurs by solving:

$$ax^2 + bx + c = 0$$

The **solutions** of $0 = ax^2 + bx + c$ are the same as the x-intercepts of $y = ax^2 + bx + c$ and are also known as the **roots** of $ax^2 + bx + c$.

Factorisation

There are two ways to solve a quadratic equation. Sometimes we can "factorise", which is the reverse of expanding brackets. If $a = 1$, then we seek two numbers that multiply to c and add to b.

Example:

$$x^2 + 4x + 3 = 0$$

What pair multiplies to 3 and adds to 4?

$$x^2 + 3x + 1x + 3 = 0$$

3 and 1 of course!

$$(x + 3)(x + 1) = 0$$

This means that either $x + 3 = 0$, so $x = -3$, or that $x + 1 = 0$, so $x = -1$. This approach isn't always possible, so the most reliable method to use (which *always* works!) is...

The Quadratic Formula

The quadratic formula:

If $ax^2 + bx + c = 0$ and $a \neq 0$ then:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{1}$$

The **discriminant** is the "bit under the square root." It indicates how many roots exist and of what type:

- $b^2 - 4ac > 0$ indicates two real and distinct roots (x_1 and x_2)
- $b^2 - 4ac = 0$ indicates real and repeated roots ($x_1 = x_2$)
- $b^2 - 4ac < 0$ indicates complex roots ($x = p \pm jq$)

Example

Let's use this method to solve the same equation as before:

$$x^2 + 4x + 3 = 0$$

In this case, the constants are $a = 1$, $b = 4$ and $c = 3$, so we substitute these into the formula:

$$\begin{aligned} x &= \frac{-4 \pm \sqrt{4^2 - 4 \times 1 \times 3}}{2 \times 1} = \frac{-4 \pm \sqrt{16 - 12}}{2} \\ &= \frac{-4 \pm \sqrt{4}}{2} = \frac{-4 \pm 2}{2} = -2 \pm 1 \end{aligned}$$

Thus, the solutions are $x = -3$ and $x = -1$, as we had found by factorisation.

Roots and Discriminants

Furthermore, if there are two distinct, real roots x_1 and x_2 to the quadratic $y = ax^2 + bx + c$, then it is possible to re-write the quadratic in the form:

$$y = a(x - x_1)(x - x_2)$$

However, if there are real and repeated roots, $x_1 = x_2$, then it is possible to re-write the quadratic in the form:

$$y = a(x - x_1)^2$$

Using MATLAB to Plot Polynomials

Using MATLAB to plot a function allows us to automate the process.

We can use different techniques to plot polynomials:

- the `plot` command - we define and plot the co-ordinates.
- the `fplot` command - we define the equation, then plot within a desired range.

Using MATLAB to Plot Polynomials

The `plot` command:

```
x=[-5:10]; %set the x range
y=5*x.^2-2*x+4; %state the equation
plot(x,y) %plot the function
```

Using MATLAB to Plot Polynomials

The `fplot` command:

```
syms y(x) %Declare the variables
y=7*x-4;  %Provide the equation
fplot(y,[0 5]) %Plot the function within a specified range.
```